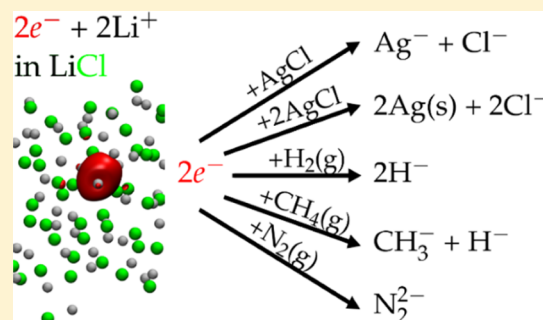


Chemistry of Solvated Electrons in Molten Alkali Chloride Salts

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Supporting Information

ABSTRACT: We use ab initio molecular dynamics to study the electronic structure and the chemistry of the solvated electrons created when alkali atoms are dissolved in alkali chloride molten salt. We find that solvated electrons pair up to form a two-electron species. This species is a very strong reductant which reduces Ag^+ to Ag^- or $\text{Ag}(\text{s})$, dissociates H_2 to form 2H^- , dissociates CH_4 to form CH_3^- and H^- , and converts N_2 to N_2^{2-} . We also study how these properties depend on the composition of the alkali chloride salt.



1. INTRODUCTION

Alkali metals dissolve in molten alkali halides and ionize spontaneously to produce solvated electrons and alkali ions. Experiments have been performed to determine the solubility, the spectroscopy, the electrical conductivity, and the phase diagram of various systems of this type.^{1–7} A valuable review of this literature, by Bredig, is available on the web.⁸

A variety of systems that catalyze many types of reactions are classified as base catalysts.^{9–13} These catalysts share two characteristics: they are Lewis bases and they catalyze reactions that proceed through a negative ion intermediate. A few examples of such catalysts are irreducible oxides pretreated at high temperature,⁹ supported alkali metals,¹³ zeolites doped with alkali metals,^{14–19} alkali hydroxides, amines tethered to solid surfaces, hydrotalcite, supported metal clusters with alkali metal promoters,^{20–22} and others.^{9–13} Examples of the type of reactions affected by the base catalysts are alkene and alkyne isomerization, alkylation reactions, Tishchenko reactions, Michael addition, Knoevenagel condensations, epoxide reactions with CO_2 to make organic carbonates, and others.^{9–13} It is believed that most (or perhaps all) base catalysts act by donating an electron to the substrate to create a negative ion intermediate which then reacts to generate products. The electron donated to the substrate is returned to the catalyst when the products are desorbed. In a manner of speaking, the electron is the catalyst. It is therefore natural to ask whether solvated electrons in molten salts might be used as base catalysts.

It has been stated^{23–27} that a solvated electron is the strongest possible reductant in any given solvent. In this article, we study the solvated electrons originating from Li dissolved in molten LiCl, Li dissolved in a molten LiCl–NaCl mixture, and Na dissolved in molten NaCl. We find that the solvated electrons pair up to form a two-electron species, whose energy fluctuates strongly over time because of the motion of the

surrounding molten salt ions. This is important in relation to base catalysis because when the two-electron species has high energy, the electrons are more likely to be transferred to the reactants. Solvated electrons could therefore be very potent base catalysts.

Rieke²⁸ has shown that alkali metals dissolved in organic solvents reduce the metal salts to make metallic clusters which have remarkable reactivity. We have investigated whether similar chemistry can be performed when the solvent is a molten salt. We find that AgCl, dissolved in molten LiCl in which Li was dissolved, will spontaneously form metal particles when there are equal amounts of Ag^+ and e^- . We also find an unexpected Ag^- ion when there are two e^- and one Ag^+ .

In the last part, we consider the reactions between the solvated electrons and gaseous H_2 , CH_4 , and N_2 to form 2H^- , $\text{CH}_3^- + \text{H}^-$, and N_2^{2-} . The reaction energies are strongly dependent on the molten salt composition, which indicates that the reactivity of the solvated electrons can be tuned by changing the nature of the molten salt.

Other possible advantages of using solvated electrons as base catalysts are enumerated in the Summary section.

2. COMPUTATIONAL METHODS

The molten salts are modeled as freestanding films with periodic boundary conditions in two directions and vacuum on both sides of the film (in the direction perpendicular to the film). Three such films have been studied: one contains 45LiCl in the unit cell, the second contains 45NaCl, and the third is a 29LiCl–16NaCl mixture. These choices were made to find out how the properties of a solvated electron change when the cation in the molten salt is changed.

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The thickness of the 45LiCl film is roughly 14 Å, that of 29LiCl–16NaCl is 15 Å, and that of 45NaCl is 19 Å. The dimensions of the cells repeated periodically in the direction parallel to the surface of the film are 12 Å × 12 Å. In the perpendicular direction, the dimension of the cell is 32 Å for 45LiCl, 34 Å for the 29LiCl–16NaCl mixture, and 35 Å for 45NaCl.

The molten salts are studied with *ab initio*, constant temperature molecular dynamics (MD),^{29–31} which is implemented in the VASP program.^{32–35} The energy in these simulations is provided by density functional theory (DFT), with a plane wave basis set (350 eV energy cutoff), and Γ -point sampling of the reciprocal space. The exchange–correlation effects are approximated by the Perdew, Burke, and Ernzerhof (PBE) functional,³⁶ and we use the D3 van der Waals correction.³⁷ The calculations use the projector augmented-wave method^{38,39} and include 1, 1, 1, 4, 5, 7, and 11 valence electrons for every Li, Na, H, C, N, Cl, and Ag atom, respectively. Spin polarization is included in all calculations; unless otherwise stated, the spin state is a singlet.

Constant temperature MD simulations are performed at 900 K for LiCl and LiCl–NaCl mixtures and at 1100 K for NaCl. These temperatures are slightly above the experimental values for the melting points of LiCl (883 K), the LiCl–NaCl mixture (845 K), and NaCl (1074 K).⁴⁰ A visual inspection of the movies (Supporting Information) showing the motion of the ions confirms that the systems are liquid. The motion of the ions is treated classically, and the time step is 1 fs.

The cumulative average energy [$U(t)$, eq 1] of a given system, which is the same as the thermodynamic internal energy, is calculated as the time average of the instantaneous total energy, which consists of the classical kinetic energy E_{kin} of the ions and the interaction energy provided by DFT

$$U(t) = \frac{1}{t - t_0} \int_{t_0}^t (E_{\text{DFT}}(\tau) + E_{\text{kin}}(\tau)) d\tau \quad (1)$$

Because the system is held at constant temperature, the total energy fluctuates as time evolves. The motion prior to time t_0 is not included in the average energy, and t_0 is chosen such that the system is sufficiently randomized to “forget” the initial conditions. For LiCl, the energy $U(t)$ became stable after $t_0 = 0.4$ ps, for NaCl after $t_0 = 2.0$ ps, and for the LiCl–NaCl mixture after $t_0 = 13$ ps. Presumably, the long time required for equilibrating LiCl–NaCl is due to the need to achieve a proper mixing of the two cations. Hence, $t_0 = 2.5$ ps is sufficient to study CH₄ and N₂ solvation in the prerun LiCl–NaCl mixture.

MD simulations are used to calculate reaction energies, that is, the energy difference between the final state and the initial state of a reaction. In the case when a gas molecule reacts with (or dissolves in) a melt, we calculate the difference between the energy of the liquid and the molecule after reaction (or dissolution) minus the energy of the pure liquid, minus the energy of the molecule in the gas phase. The energies of the gas-phase molecules are calculated as follows: The center of mass of the molecule is fixed, but the molecule can rotate and vibrate. Therefore, the calculation delivers the electronic, classical vibrational, and classical rotational energy of the molecule in vacuum. To this, we add the translational energy $3k_{\text{B}}T/2$ (k_{B} is the Boltzmann constant).

The Bader charges^{41–43} of various species are calculated at 1 ps intervals, and the average charge and its standard deviation

(because of the movement of the ions) are reported. In this study, e is the proton charge.

3. RESULTS AND DISCUSSION

3.1. Solvation of Li in LiCl. In this section, we investigate the dissolution of one Li atom and two Li atoms in a molten 45LiCl film. Figure 1 shows the cumulative energy for these two systems.

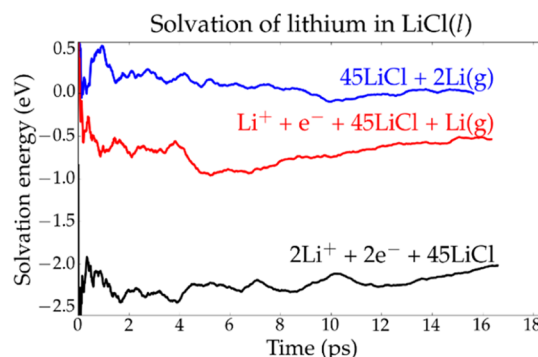
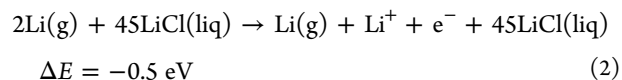


Figure 1. Cumulative mean energy as a function of time ($t - t_0$) for one Li atom dissolved in the 45LiCl melt (red line) and two Li atoms dissolved in the 45LiCl melt (black line). The blue line shows the energy of $2\text{Li(g)} + 45\text{LiCl(liq)}$. The longtime energy value of $2\text{Li(g)} + 45\text{LiCl(liq)}$ is the reference energy (it is set equal to zero).

As the graph contains cumulative averages, we use the energy values obtained at the longest times to judge the stability of different systems. In the calculations reported in Figure 1, the longtime energy of $2\text{Li(g)} + 45\text{LiCl(liq)}$ was set equal to zero, and this is the reference energy for the red and black curves. For example, the red curve shows the difference between the energy of the system $\text{Li(g)} + \text{Li}^+ + \text{e}^- + 45\text{LiCl(liq)}$ and the energy of $2\text{Li(g)} + 45\text{LiCl(liq)}$.

When inserted into the 45LiCl(liq) melt, a Li atom ionizes rapidly to form a Li^+ ion and a solvated electron. This Li^+ ion is indistinguishable from the other Li ions present in the melt. This process is represented by the equation



In all equations of this kind in what follows, it is understood that the ion and the electron are dissolved in the molten salt. The transfer of a Li atom from the gas into the melt stabilizes the total energy by -0.5 eV (Figure 1 and Table 1). This energy is the difference between the red curve and the blue curve, at the longest simulation time. Because we perform constant temperature MD simulations, the energy of each system fluctuates. The longtime values of the mean energy are the most reliable, because one has better statistics for the cumulative average. The electron produced by Li ionization forms a cavity in the melt. The solvated electron is detached from and is uncorrelated with the Li ion formed by the ionization of the Li atom. The cavity is surrounded by moving Li ions.

When two Li ions are taken from the gas and placed into the melt, they ionize spontaneously in a process represented by

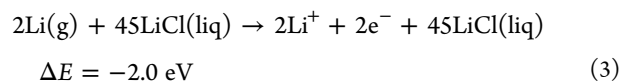


Table 1. Solvation and Reaction Energies

reaction (900 K)	ΔE (eV)	reaction (1100 K)	ΔE (eV)
Alkali Metal Solvation in Alkali Chlorides			
$\text{Li(g)} + 45\text{LiCl} \rightarrow \text{Li}^+ + \text{e}^- + 45\text{LiCl}$	-0.5	$\text{Na(g)} + 45\text{NaCl} \rightarrow \text{Na}^+ + \text{e}^- + 45\text{NaCl}$	-0.9
$2\text{Li(g)} + 45\text{LiCl} \rightarrow 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$	-2.0	$2\text{Na(g)} + 45\text{NaCl} \rightarrow 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$	-1.9
$8\text{Li(g)} + 45\text{LiCl} \rightarrow 8\text{Li}^+ + 8\text{e}^- + 45\text{LiCl}$	-8.5		
$2\text{Li(g)} + 29\text{LiCl} + 16\text{NaCl} \rightarrow 2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$	-2.4		
Ag^+ Reacting with Alkali Metal/Alkali Chlorides			
$\text{AgCl(g)} + 45\text{LiCl} \rightarrow \text{Ag}^+ + \text{Cl}^- + 45\text{LiCl}$	-1.8	$\text{AgCl(g)} + 45\text{NaCl} \rightarrow \text{Ag}^+ + \text{Cl}^- + 45\text{NaCl}$	-2.4
$\text{AgCl(g)} + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow \text{Ag}^- + \text{Li}^+ + 46\text{LiCl}$	-4.7	$\text{AgCl(g)} + 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} \rightarrow \text{Ag}^- + \text{Li}^+ + 46\text{NaCl}$	-4.1
$2\text{AgCl(g)} + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow \text{Ag}_2 + 47\text{LiCl}$	-7.4	$2\text{AgCl(g)} + 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} \rightarrow \text{Ag}_2 + 47\text{NaCl}$	-7.0
$2\text{AgCl(g)} + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow 2\text{Ag(s)} + 47\text{LiCl}$	-10.6		
H_2, N_2, and CH_4 Reacting with Alkali Metal/Alkali Chlorides			
$\text{H}_2(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow 2\text{H}^- + 2\text{Li}^+ + 45\text{LiCl}$	-2.3	$\text{H}_2(\text{g}) + 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} \rightarrow 2\text{H}^- + 2\text{Na}^+ + 45\text{NaCl}$	-0.9
$\text{CH}_4(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow \text{CH}_3^- + \text{H}^- + 2\text{Li}^+ + 45\text{LiCl}$	-0.7	$\text{CH}_4(\text{g}) + 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} \rightarrow \text{CH}_3^- + \text{H}^- + 2\text{Na}^+ + 45\text{NaCl}$	+0.9
$\text{N}_2(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow \text{N}_2^{2-} + 2\text{Li}^+ + 45\text{LiCl}$	-1.7	$\text{N}_2(\text{g}) + 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} \rightarrow \text{N}_2^{2-} + 2\text{Na}^+ + 45\text{NaCl}$	unstable, N_2 desorbs
$\text{N}_2(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow \text{N}_2^{2-} + 2\text{Li}^+ + 45\text{LiCl}$ (triplet)	-1.5	$\text{N}_2(\text{g}) + 2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} \rightarrow \text{N}_2^{2-} + 2\text{Na}^+ + 45\text{NaCl}$ (triplet)	0.0
$\text{CH}_4(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl} \rightarrow \text{CH}_3^- + \text{H}^- + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$	-0.4		
$\text{N}_2(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl} \rightarrow \text{N}_2^{2-} + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$	unstable, N_2 desorbs		
$\text{N}_2(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl} \rightarrow \text{N}_2^{2-} + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$ (triplet)	-1.1		

The evolution of the energy of $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ is shown by the black curve in Figure 1. This is referenced to the blue curve of the same figure. If the two solvated electrons produced by this process were spatially separated and did not interact with each other, dissolving two Li atoms in the melt would stabilize the energy by 1 eV (0.5 eV per electron, according to eq 2). However, adding two Li atoms in the melt stabilizes the energy by -2.0 eV. This indicates that the electrons form a combined chemical species. Indeed, the simulations show that the two electrons occupy the same cavity in the liquid and form a singlet spin state. We call this species a “bipolaron” and note that bipolaron formation was reported in previous simulations for Na in NaCl(liq).⁴⁴

The measured activity coefficient in dilute solutions of Li in LiCl is small, and Liu and Poignet⁴⁵ suggested that this is due to the formation of a compound through the association of Li with LiCl. The presence of Li_8 clusters has been reported⁴⁶ in a solution of roughly 1 mol % Li in LiCl–3 wt % Li_2O . Therefore, we examined the behavior of a Li_8 cluster introduced into the molten LiCl film. We found that the Li_8 cluster quickly breaks apart and forms four bipolarons and eight Li ions (see the Supporting Information for details).

3.2. Electronic Properties of Li–LiCl. Figure 2 gives information about the electronic structure of the $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ system. Figure 2a shows the time evolution of the eigen energy for several orbitals. The observed oscillations are caused by the thermal motion of the ions. The blue graph (LUMO) shows the energy of the lowest unoccupied orbital of the salt (the “conduction” band) and the green graph (HOMO – 1) shows the top of the “valence band” of the salt. As usual, the band gap provided by DFT is too small. Of more interest to us are the two degenerate spin–orbitals in the band gap, one with spin up [HOMO(+)] and the other with spin down [HOMO(–)]. These are the orbitals occupied by the two electrons forming the bipolaron. The energy of these orbitals is

a proxy for the ability of $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ to donate electrons. As base catalysts act through the formation of a negative ion intermediate, the higher the energy of the electron-donating orbitals, the more likely is the formation of such an intermediate. The HOMO energy varies significantly with time because of the ionic motion in the melt; the standard deviation of the orbital energies is 0.36 eV. The difference between the highest energy and the lowest energy, of the same orbital, is close to 1 eV. Therefore, the ability of the bipolaron to donate an electron changes with time because of the continuous changes in its environment. This is an important distinction between the bipolarons in a melt and the F-centers in solids; in the latter, the ions vibrate but do not travel.

A surface of equal charge density of the electrons in the bipolaron is shown in Figure 2b. The pictures were made at times $t - t_0 = 3.6$ and 15.6 ps. A long series of such pictures, taken at different times, is given in Figure S1 (Supporting Information). The snapshots show that most of the time the bipolaron is localized in a single cavity and that the cavity is surrounded by Li^+ ions. Because of the thermal motion of the ions, the shape of the equal charge density surface of the orbital changes continuously. Occasionally, the bipolaron jumps out of its cavity to a new cavity, where it is localized for a while. An example of such a jump happens at $t - t_0 = 7.0$ ps, when, for about ~ 0.1 ps, the bipolaron is localized simultaneously in two cavities having comparable electron densities, after which both electrons are localized in the new cavity (Figure S1, Supporting Information).

The solvation energy of the dissolved alkali atoms is determined by the ionization energy of the atom (in gas phase) and the solvation energies of the electrons and the ions formed by ionization. The latter quantities cannot be determined separately; we can only determine their sum. To assess the stability of the solvated electrons independently of the solvation of the alkali ion, we calculated the ionization

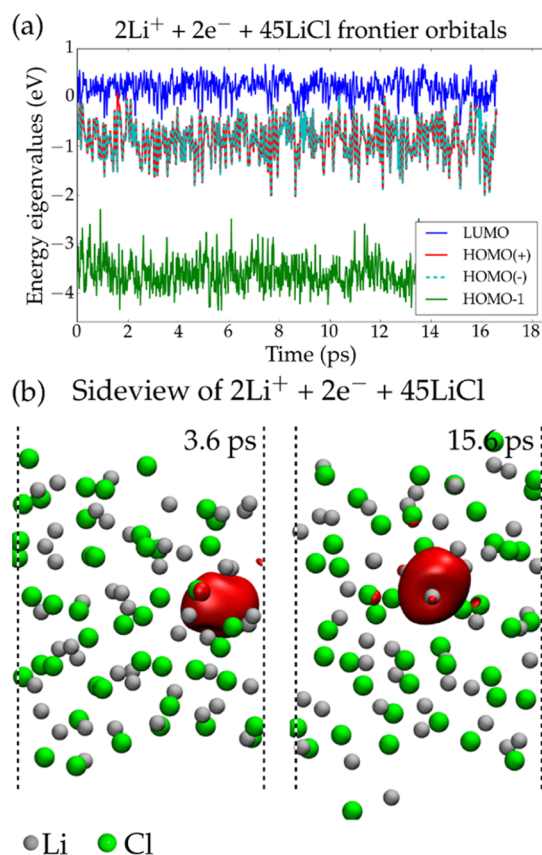


Figure 2. (a) Time evolution of the frontier orbital energies in $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$. Blue shows the energy of the bottom of the conduction band (LUMO), green shows the top of the valence band (HOMO – 1), and red and dashed cyan show the energies of the bipolaron orbitals [HOMO(+) and HOMO(–)]. (b) Atomic configuration of $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ at 3.6 and 15.6 ps. In red color, we show the surface $|\psi_{\text{HOMO}(+)}(\mathbf{r})|^2 + |\psi_{\text{HOMO}(-)}(\mathbf{r})|^2 = 0.02 \text{ \AA}^{-3}$, where $\psi_{\text{HOMO}(\pm)}(\mathbf{r})$ are the orbitals occupied by the electrons in the bipolaron.

energy⁴⁷ (IE) of the melts containing a bipolaron. IE varies in time and was determined from the atomic configurations taken at 1 ps intervals along the MD trajectories. For the IE calculations, we increased the computational cell to 50 Å in the direction (z) perpendicular to the molten salt film, applied a dipole correction in the z direction, and averaged the resulting electrostatic potentials in the directions parallel to the film (x and y). The molten salt film has vacuum on both sides and therefore has two surfaces. At any given time, the energy to remove the electron through the upper surface differs from the energy to remove it through the lower surface. This happens because the ions at the two faces are arranged differently. We use the lower of the two values at each point in time, report the average value and the standard deviation in Table 2, and find IE to be 3.03 ± 0.54 eV for $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$. For comparison, the IE of Li metal is 2.9 eV.⁴⁸ Table 2 also shows the energy for the removal of an electron from $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$ and from $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$. A higher stability of the bipolaron in LiCl (higher IE) is to be expected, given the smaller radius of Li^+ than that of Na^+ . However, it is surprising that the binary system ($2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$) has the lowest IE of the three molten salts.

The large values of the standard deviation in IE (Table 2) indicate that there are very large fluctuations in the work

Table 2. Mean IE $\pm$ Standard Deviation (the Energy to Move One Electron from the Melt to Infinity) Averaged Over Around 16 Configurations Selected at Intervals of 1 ps^a

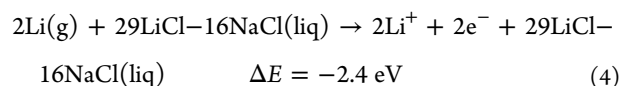
melt composition	IE (eV)
$2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$	3.03 ± 0.54
$2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$	2.77 ± 0.40
$2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$	2.89 ± 0.44

^aThe energy to remove the electron through the upper face of the film differs from the energy of removing it through the lower face. We use the lower of the two values at each point in time.

function, which are consistent with the large fluctuations in the HOMO orbital energies (Figure 2a). These fluctuations are very rapid (<1 ps timescale), and any experimental measurement will determine the time average. However, they are very important if one attempts to use the electrons for a reduction reaction, as electron transfer will be affected by the orbital energies. Given the large standard deviation in IE, we conclude that the fluctuations in the binding energy of the electron are such that, roughly, half of the time the IE of the $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ melt is more facile than the ionization of solid Li. The melt could therefore be a better electron donor than metallic Li.

Alkali metals supported on oxides have been used as base catalysts^{9,10,13} for many classes of reactions because they are good electron donors (Lewis bases). The ionization energies calculated here show that the solvated electrons produced when an alkali is dissolved in a molten salt are likely to catalyze the same reactions as supported alkali metals. The difference in the IE of solvated electrons, in the melts of different compositions, shows that the catalytic activity of the solvated electron can be tuned by changing the nature of the molten salt. If the melt is a mixture of two halides, an additional tuning of the electron properties can be achieved by varying the salt composition. The stability of the ionic species formed by the reactions with the solvated electrons also is significantly affected by the molten salt composition.

3.3. Solvation of Li in LiCl–NaCl Mixture. One of the goals of this paper is to study how the chemistry of solvated electrons can be modified by using a binary solution of molten salts. Binary solutions are particularly convenient if one needs to use solvated electrons to perform chemistry at lower temperature because mixtures melt at substantially lower temperatures than their pure components.^{40,49} Because of this, we examined the dissolution of two Li atoms into a molten mixture containing 29LiCl and 16NaCl. The reaction



is exoergic and produces -1.2 eV per dissolved Li atom (Figure 3a). The increased stabilization of $2\text{Li}^+ + 2\text{e}^-$ in the binary molten salt compared to pure LiCl (-2.4 eV compared to -2.0 eV) indicates that more Li can be solvated in a LiCl–NaCl mixture than in a pure LiCl molten salt. The solubility of Li in LiCl is quite low, and experiments have found 0.45 mol % Li solubility in LiCl(l) at 881 K⁵⁰ and 0.5 ± 0.2 mol % at 913 K.⁵

In $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl}-16\text{NaCl}$, the two electrons, donated by the two Li atoms, form a bipolaron. The bipolaron is located close to the surface of the melt (Figure 3b), the

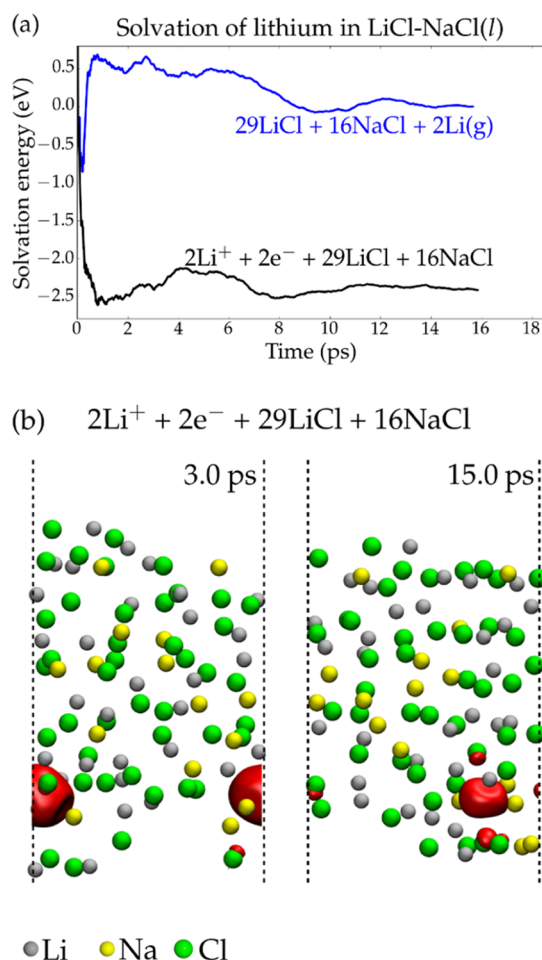
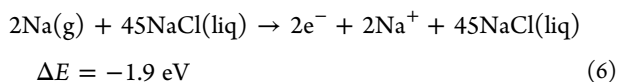
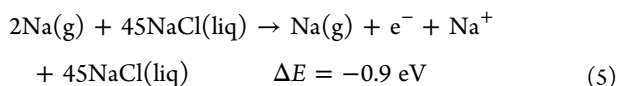


Figure 3. (a) Blue curve is the energy of $2\text{Li}(\text{g}) + 29\text{LiCl} - 16\text{NaCl}$. Zero energy is the cumulative mean energy of this system after 16 ps. The black curve is the cumulative mean energy of $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} - 16\text{NaCl}$. The energy difference between the two curves is the energy of $2\text{Li}(\text{g}) + 29\text{LiCl} + 16\text{NaCl} \rightarrow 2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$. (b) Atomic configuration of $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$ after $t - t_0 = 3.0$ ps and $t - t_0 = 15.0$ ps. In red color, we show the constant density surface $|\psi_{\text{HOMO}(+)}(\mathbf{r})|^2 + |\psi_{\text{HOMO}(-)}(\mathbf{r})|^2 = 0.02 \text{ \AA}^{-3}$.

position of which is different from that of the bipolaron in LiCl (Figure 2b).

3.4. Solvation of Na in Molten NaCl. Bipolarons are also formed when 2Na atoms are dissolved in molten 45NaCl. Details about the sodium–sodium chloride systems are given in the Supporting Information. Here, we note that

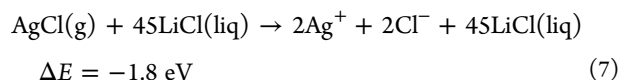


The energy of formation of two noninteracting solvated electrons, for 2Na dissolved in NaCl(liq), is -1.8 eV (-0.9 eV multiplied by 2), whereas the energy of the system in which a bipolaron is formed is -1.9 eV . Therefore, in the sodium–sodium chloride system, the bipolaron formation energy, from the two noninteracting solvated electrons, is -0.1 eV , which is within the errors in the simulations. A consequence of the small energy difference between the bipolaron and the two

noninteracting solvated electrons is that the bipolaron electrons occasionally move apart. This is seen at time $t - t_0 = 13 \text{ ps}$ in Figure S3c in the Supporting Information, which shows the time evolution of the HOMO(+) and HOMO(−) eigen energies in $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$.

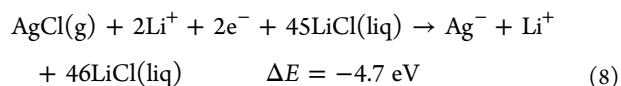
3.5. Reactions between AgCl and Li/LiCl. Alkali metals are strong reducing agents and have been used to prepare very reactive metal particles suspended in organic solvents.²⁸ We hope that a similar reduction might produce interesting metal nanoparticle catalysts or exotic metal ions in the molten salts. Here, we report the reduction of Ag^+ ions by solvated electrons in molten LiCl. Four cases are investigated: (1) An AgCl molecule is added to the 45LiCl melt to test whether it is dissolved and ionized. (2) An AgCl molecule is added to $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ to study what happens when there are more solvated electrons than Ag ions, and we find that an Ag^- ion is formed. (3) Two AgCl molecules are added to $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$, resulting in one e^- per Ag^+ ion and the formation of an Ag_2 species. (4) We calculate whether the Ag_2 species formed in step (3) would be more stable than the two atoms in bulk solid Ag; we find that energy favors the formation of solid Ag. In all these calculations, the energy of two isolated AgCl molecules in the gas phase is used as reference. The energies of these processes are given in Table 1 and in Figure 4a.

Moving one AgCl molecule from the gas phase into the 45LiCl melt stabilizes the energy



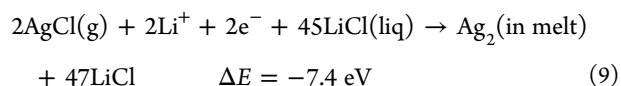
In the molten 45LiCl solvent, AgCl dissociates into ions. The Bader charge on Ag^+ is $+0.49 \pm 0.03e$ (Table 3). The large energy gain is likely due to the solvation energy of the ions. It is interesting to note that AgCl is not soluble in water.

When one AgCl molecule is added to $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$, the reaction is



The formation of Ag^- is probably aided by the fact that this ion is better solvated than the bipolaron electrons. The Bader charge on Ag^- is $-0.75 \pm 0.04e$ (Table 3), and the Ag^- ion is surrounded by Li^+ ions (Figure 4b) which causes a large solvation energy. The electrons from the bipolaron have been transferred to the 5s-state of Ag whose energy is in the band gap of the molten salt. This electron transfer converts Ag^+ to Ag^- . The shape of the 5s orbital is not spherical and is distorted toward the Li^+ ions situated close to Ag^- , as shown by the constant electron density plot of the two electrons (red surface in Figure 4b).

In the previous calculation, the system had two solvated electrons and one Ag atom. This excess of electrons causes the formation of the Ag^- ion. Next, we examine what happens when two AgCl molecules are added to $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$. The two solvated electrons reduce both Ag ions to form a neutral Ag_2 species



The driving force of -7.4 eV for this reduction is very high, emphasizing the strong reducing nature of the solvated

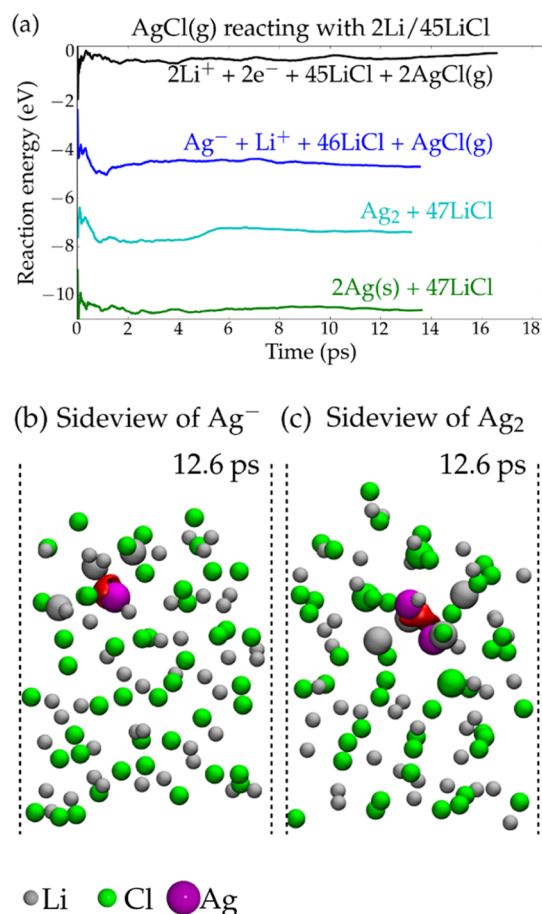


Figure 4. (a) Mean solvation energies as a function of time ($t - t_0$) for one AgCl molecule (blue) and two AgCl molecules (cyan) dissolved in $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$. The reference energy is that of two AgCl in the gas phase and $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ (0 eV, black). Also included is the reaction energy of $2\text{AgCl}(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl} \rightarrow 47\text{LiCl} + 2\text{Ag}(\text{s})$ (green). Side view of (b) $\text{Ag}^- + \text{Li}^+ + 46\text{LiCl}$ at $t - t_0 = 12.6$ ps and (c) $\text{Ag}_2 + 47\text{LiCl}$ at $t - t_0 = 12.6$ ps. The Li and Cl atoms within 3.0 Å of the Ag atoms are enlarged. In red color, we show the constant density surface $|\psi_{\text{HOMO}(+)}(\mathbf{r})|^2 + |\psi_{\text{HOMO}(-)}(\mathbf{r})|^2 = 0.08 \text{ Å}^{-3}$.

electrons. However, the reaction energy is less than twice the reaction energy of Ag^- formation (eq 8). This suggests that by adjusting the ratio of solvated electrons and Ag^+ , we can form negatively charged Ag clusters. This is important because if all Ag clusters were neutral, they would aggregate to form bulk Ag metal. Figure 4c shows that the Ag_2 species is surrounded by both Li^+ and Cl^- ions and that the electrons from the bipolaron form a covalent bond between the Ag atoms, that is, the constant electron density surface of the HOMO orbitals is situated between the Ag atoms (red surface in Figure 4c). The Bader charge on Ag_2 is $-0.19 \pm 0.08e$.

Finally, we calculate the energy of taking out Ag_2 and forming two Ag units in a bulk Ag fcc crystal. We use the heat capacity of solid Ag to account for the energy to heat it from 0 to 900 K (section S5 in the Supporting Information). The energy of $47\text{LiCl} + 2\text{Ag}(\text{s})$ is -10.6 eV, which is more stable than that of $2\text{AgCl}(\text{g}) + 2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$; hence, the calculation indicates that it is energetically favorable to form larger metallic clusters of solid Ag if the Ag to solvated electron ratio is 1:1. However, if the amount of Li exceeds that of Ag, the system will likely make negatively charged Ag clusters. It is

Table 3. Bader Charges on Product Species

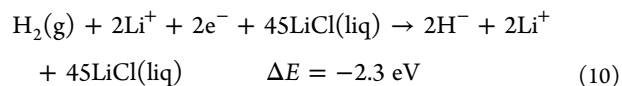
system	species	Bader charge (e)	species	Bader charge (e)
2Li/45LiCl Systems				
$\text{Ag}^+ + \text{Cl}^- + 45\text{LiCl}$	Ag^+	$+0.49 \pm 0.03$		
$\text{Ag}^- + \text{Li}^+ + 46\text{LiCl}$	Ag^-	-0.75 ± 0.04		
$\text{Ag}_2 + 47\text{LiCl}$	Ag_2	-0.19 ± 0.08		
$2\text{H}^- + 2\text{Li}^+ + 45\text{LiCl}$	H^-	-0.81 ± 0.01	H^-	-0.82 ± 0.01
$\text{CH}_3^- + \text{H}^- + 2\text{Li}^+ + 45\text{LiCl}$	CH_3^-	-0.85 ± 0.01	H^-	-0.82 ± 0.01
$\text{N}_2^{2-} + 2\text{Li}^+ + 45\text{LiCl}$	N_2^{2-}	-1.62 ± 0.04		
$\text{N}_2^{2-} + 2\text{Li}^+ + 45\text{LiCl}$ (triplet)	N_2^{2-}	-1.66 ± 0.02		
2Li/29LiCl + 16NaCl Systems				
$\text{CH}_3^- + \text{H}^- + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$	CH_3^-	-0.84 ± 0.02	H^-	-0.80 ± 0.02
$\text{N}_2^{2-} + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$ (triplet)	N_2^{2-}	-1.47 ± 0.23		
2Na/45NaCl Systems				
$\text{CH}_3^- + \text{H}^- + 2\text{Na}^+ + 45\text{NaCl}$	CH_3^-	-0.79 ± 0.02	H^-	-0.73 ± 0.02
$\text{N}_2^{2-} + 2\text{Na}^+ + 45\text{NaCl}$ (triplet)	N_2^{2-}	-0.41 ± 0.17		
$2\text{H}^- + 2\text{Na}^+ + 45\text{NaCl}$	H^-	-0.73 ± 0.02	H^-	-0.73 ± 0.02

possible that an electrostatic repulsion between such clusters will prevent their coarsening. The catalytic properties of such negatively charged clusters have not been explored.

Given the use of alkali to reduce salts to metals in many industrial processes, we assume that practically all halides dissolved in an alkali halide melt containing dissolved Li will be reduced to form metal clusters. This may provide a route for preparing metal nanoparticle catalysts.

The reactions between AgCl and Na/NaCl are discussed in detail in section S2.3 in the Supporting Information. We find the same qualitative results as for Li/LiCl: one AgCl molecule added to $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ results in the formation of Ag^- , whereas two AgCl molecules added to $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ result in the formation of an Ag_2 dimer. The main difference between NaCl and LiCl is that AgCl has lower energy in 45NaCl (-2.4 eV) than in 45LiCl (-1.8 eV), whereas AgCl has higher energy in $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ (-4.1 eV for one AgCl unit and -7.0 eV for two AgCl units) than in $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ (-4.7 eV for one AgCl unit and -7.4 eV for two AgCl units). This means that the solvated electrons in molten NaCl are weaker reducing agents than the solvated electrons in molten LiCl. Because the solvated electrons in LiCl are more stable (have higher IE, Table 2) than in NaCl, the main difference in reduction strength must come from Li^+ being better than Na^+ at stabilizing the negative reaction products.

3.6. Reactions of Li/LiCl with H_2 , CH_4 , or N_2 . At this point, we expect that the solvated electrons are very reactive and have therefore examined the reactions between $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}(\text{liq})$ and $\text{H}_2(\text{g})$, $\text{CH}_4(\text{g})$, or $\text{N}_2(\text{g})$. The time evolution of various reaction energies is given in Figure 5a. The zero on the energy scale is the average energy of $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}(\text{liq})$ after 16 ps and the energy of H_2 , CH_4 , or N_2 in the gas phase. The red curve shows the time evolution of the energy of the reaction



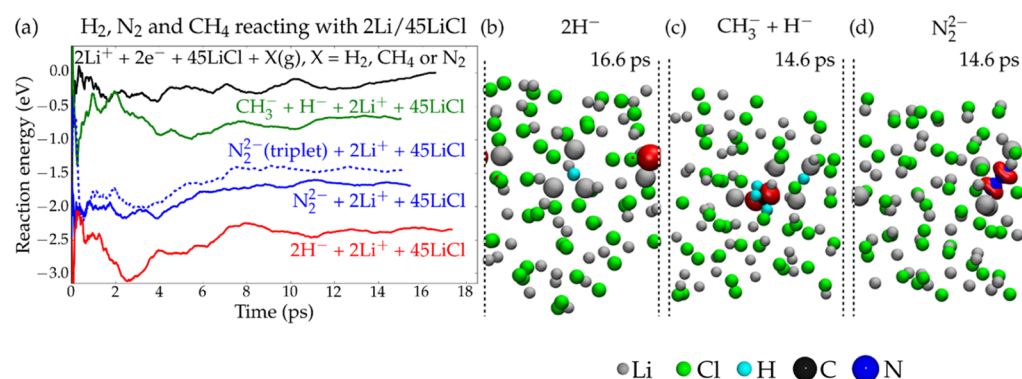


Figure 5. (a) Mean energies as a function of elapsed time ($t - t_0$) for the dissociation of $\text{H}_2(\text{g})$, dissociation of $\text{CH}_4(\text{g})$, and for the formation of N_2^{2-} from $\text{N}_2(\text{g})$, all dissolved in $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$. Side view of (b) $2\text{H}^- + 2\text{Li}^+ + 45\text{LiCl}$ after $t - t_0 = 16.6$ ps, (c) $\text{H}^- + \text{CH}_3^- + 2\text{Li}^+ + 45\text{LiCl}$ after $t - t_0 = 14.6$ ps, and (d) $\text{N}_2^{2-} + 2\text{Li}^+ + 45\text{LiCl}$ in singlet spin state after $t - t_0 = 16.6$ ps. The Li and Cl atoms within 2.3 Å of the solute are enlarged. In red color, we show the constant density surface $|\psi_{\text{HOMO}(+)}(\mathbf{r})|^2 + |\psi_{\text{HOMO}(-)}(\mathbf{r})|^2 = 0.10 \text{ Å}^{-3}$.

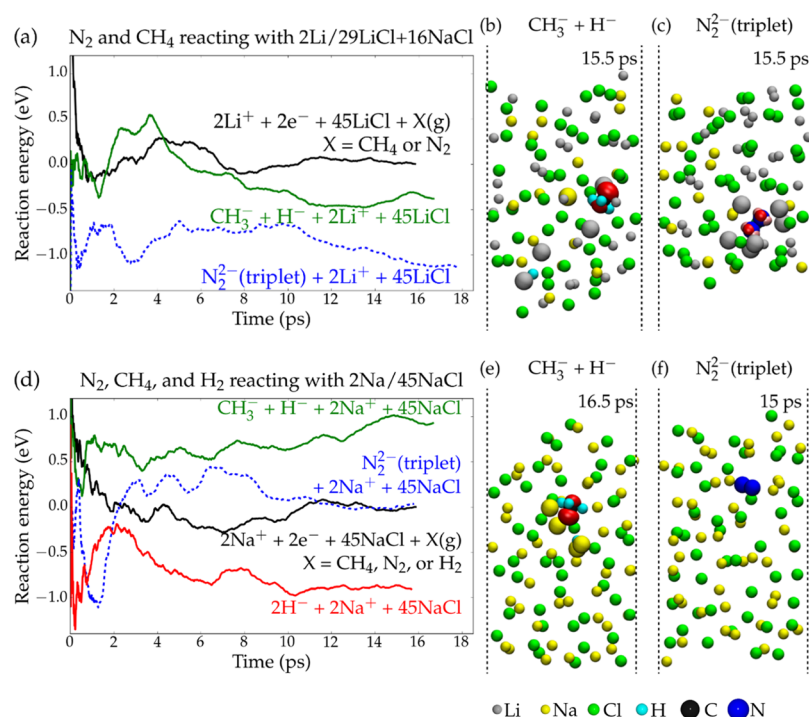


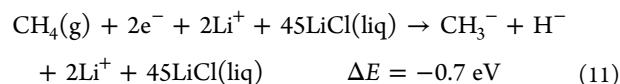
Figure 6. (a) Mean solvation energies as a function of elapsed time ($t - t_0$) for dissociated CH_4 and N_2 solvated in $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$ and compared to the molecules in the gas phase and $29\text{LiCl} + 16\text{NaCl}$ (0 eV). Side view of (b) $\text{H}^- + \text{CH}_3^- + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$ after $t - t_0 = 15.5$ ps and (c) $\text{N}_2^{2-} + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$ in triplet spin state after $t - t_0 = 15.5$ ps. (d) Mean solvation energies as a function of elapsed time ($t - t_0$) for dissociated CH_4 , nondissociated N_2 , and dissociated H_2 solvated in $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ and compared to the molecules in the gas phase and $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ (0 eV). Side view of (e) $\text{H}^- + \text{CH}_3^- + 2\text{Na}^+ + 45\text{NaCl}$ after $t - t_0 = 16.5$ ps and (f) $\text{N}_2^{2-} + 2\text{Na}^+ + 45\text{NaCl}$ in triplet spin state after $t - t_0 = 15.0$ ps. All Li, Na, and Cl atoms within 2.3 Å of the solutes are enlarged in (b,c,e,f). The constant density surface $|\psi_{\text{HOMO}(+)}(\mathbf{r})|^2 + |\psi_{\text{HOMO}(-)}(\mathbf{r})|^2 = 0.10 \text{ Å}^{-3}$ is shown in red color in (b,e). The constant density surfaces of $|\psi_{\text{HOMO}(+)}(\mathbf{r})|^2 = 0.10 \text{ Å}^{-3}$ and $|\psi_{\text{HOMO}-1(+)}(\mathbf{r})|^2 = 0.10 \text{ Å}^{-3}$ are shown in red and light red color in (c). In (f), the electron density of $\text{HOMO}(+)$ and $\text{HOMO} - 1(+)$ is less than 0.10 Å^{-3} everywhere.

One can think of the final state as being a solution of lithium hydride in the molten LiCl. Lithium hydride is a stable ionic hydride, which forms eutectics with a large number of salts. It is therefore reasonable to expect that it will mix well with the $45\text{LiCl}(\text{liq})$ melt. The large reaction energy is surprising in view of the fact that the hydrogen–hydrogen bond in H_2 must be broken. We speculate that this large reaction energy indicates that the energy of solvation of the two hydride ions is much larger than the energy of solvation of the bipolaron. This is consistent with the diffuse charge of the bipolaron compared

to that of the H^- ions. In this simulation, the two H atoms were placed far apart in the melt. The two electrons forming the bipolaron migrated quickly to the H atoms to form the hydride ions having Bader charges of $-0.81e$ and $-0.82e$ (Table 3). The H^- ions are surrounded by several Li^+ ions (Figure 5b). At the time $t - t_0 = 16.6$ ps shown in Figure 5b, the constant electron density plot (red) of the HOMO electrons shows that they are situated in the 1s state of one of the H atoms. The other H atom also has electrons occupying

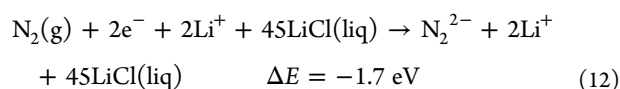
its 1s state, but the state is mixed with the valence band of the molten salt.

The green curve in Figure 5a shows the evolution of the energy of the reaction



The Bader charge on CH_3^- is $-0.85e$ and that on H^- is $-0.82e$. The CH_3^- and the H^- ions are both surrounded by Li^+ ions (Figure 5c). The solvation energy of the negative ions plays an important role in lowering the energy of CH_4 dissociation. It is challenging to find efficient catalysts that break the C–H bond of methane at the temperatures used here.⁵¹ The reaction is exoergic mainly because the negative ion products are solvated. Figure 5c shows a constant electron density plot from the HOMO orbitals, which shows that the electron occupies a p state on the carbon atom.

The addition of N_2 molecule to $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ resulted in the reaction



where the two electrons transferred to N_2^{2-} form a singlet (Figure 5d). The reaction energy for the triplet is -1.5 eV . The Bader charge of N_2^{2-} is $-1.62e$ for the singlet and $-1.66e$ for the triplet. The charge difference between the two species is below the accuracy with which the Bader charge is determined. Figure 5d shows a constant electron density plot from the HOMO orbitals, and it is seen that the two electrons form a π bond on the N_2^{2-} species.

The calculations in this section show that the bipolaron reacts with H_2 and CH_4 , causing them to dissociate and form solvated negative ions. N_2 reacts with the bipolaron to form N_2^{2-} . Because these calculations take a very long time ($\sim$ a month on 32 CPUs for one 18 ps run), we have not examined the reactivity of H^- , CH_3^- , or N_2^{2-} . We speculate that H^- and CH_3^- will react vigorously with Lewis acids. It is also conceivable that if the system had more than two electrons, N_2 might dissociate into ionic species.

3.7. Reactions of CH_4 and N_2 with Solvated Electrons in LiCl–NaCl and NaCl. To investigate the reactivity of the solvated electrons in a molten salt of a different composition, we examined the interaction of CH_4 and N_2 with $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl}(\text{liq}) + 16\text{NaCl}(\text{liq})$ and H_2 , CH_4 , and N_2 with $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}(\text{liq})$. Figure 6 summarizes the reactions between the gaseous species and the two melts, and the reaction energies are included in Table 1.

To study the energetics of CH_4 dissociation by $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$, we added a H atom and a CH_3 species into the melt. The electrons from the bipolaron moved very rapidly to form H^- (Bader charge of $-0.80 \pm 0.02e$) and CH_3^- (Bader charge of $-0.84 \pm 0.02e$, Table 3). The energy of the reaction is $\Delta E = -0.4 \text{ eV}$ (Figure 6a). The ionic dissociation products, CH_3^- and H^- , are predominantly solvated by the Li^+ ions (Figure 6b and section S4.2 in the Supporting Information).

A N_2 molecule was added to $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$ and MD simulation was performed for both a singlet and a triplet spin state. In both cases, the bipolaron electrons move to N_2 , forming N_2^{2-} . However, in the simulation in which the spin state was a singlet, the N_2 eventually desorbed from the molten salt to form $\text{N}_2(\text{g})$ and a bipolaron with a singlet spin.

The increase of entropy upon N_2 desorption will lower the free energy of the gas and increase the driving force for desorption.

The N_2^{2-} species in the triplet spin state stays solvated throughout the MD simulation and has a solvation energy of -1.1 eV . The N_2^{2-} is predominantly solvated by Li^+ ions (Figure 6c and section S4.1 in the Supporting Information). In the triplet spin state, the two unpaired electrons occupy two different π orbitals on N_2^{2-} that point in perpendicular directions (red and light red constant electron density plots in Figure 6c). It is possible that N_2^{2-} with a triplet spin does not desorb because that would leave behind an unfavorable triplet spin bipolaron.

The results above show that the solvation energy of N_2 and of CH_4 is smaller in the LiCl–NaCl mixture than in LiCl (Table 1).

The energy of CH_4 reacting with $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ to form $\text{CH}_3^- + \text{H}^- + 2\text{Na}^+ + 45\text{NaCl}$ is $+0.9 \text{ eV}$, and the energy of N_2 reacting with $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ to form $\text{N}_2^{2-}(\text{triplet}) + 2\text{Li}^+ + 29\text{LiCl} + 16\text{NaCl}$ is 0.0 eV (Figure 6d, Table 1). These reactions are therefore significantly less exoergic than the reactions of CH_4 and N_2 with $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$ or $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$ (Table 1). The lower reaction energies are caused by a weaker stabilization of the negative product ions in NaCl, which correlates with the lower Bader charge on the product ions (Table 3) and lower number of alkali ion neighbors (Figure 6e,f, and section S4 in the Supporting Information).

The Bader charge on $\text{N}_2^{2-}(\text{triplet})$ in 45NaCl is only $-0.41 \pm 0.17e$, and this is because the bipolaron electrons sometimes move away from N_2^{2-} and form solvated electrons in a triplet spin state instead. This is the case at $t - t_0 = 15 \text{ ps}$ (Figure 6f), in which the constant electron density plots (with $|\psi_{\text{HOMO}+}(\mathbf{r})|^2$ and $|\psi_{\text{HOMO}-1+}(\mathbf{r})|^2$ equal to $0.10 \text{ \AA}^{-3}$) of the unpaired orbitals do not show any orbitals on the N_2 species.

Finally, Figure 6d includes the energy of the reaction $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl} + \text{H}_2(\text{g}) \rightarrow 2\text{H}^- + 2\text{Na}^+ + 45\text{NaCl}$ (red curve). The reaction energy is -0.9 eV . Solid NaH decomposes thermally at $\sim 700 \text{ K}$,⁵² which gives some indication that the species with lower solvation energy (i.e., CH_4 solvated in $2\text{Li}^+ + 2\text{e}^- + 45\text{LiCl}$, $2\text{Li}^+ + 2\text{e}^- + 29\text{LiCl} + 16\text{NaCl}$, and $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$ and N_2 solvated in $2\text{Na}^+ + 2\text{e}^- + 45\text{NaCl}$) might only have short lifetimes before desorbing again. However, in those cases, the solvated electrons can function as catalysts if the lifetime is long enough to allow for the reactions between CH_3^- , H^- , or N_2^{2-} and other reactants.

The calculations in this section highlight how changing the molten salt cation composition modifies the chemistry of solvated electrons. However, it might be possible to further modify the reactivity by changing the anion composition from Cl^- to F^- , Br^- , or I^- . The influence of the anion can be seen in computational studies for H_2O solvated in molten LiI and molten LiCl. In both solvents, the H_2O molecule binds closely to the Li^+ ions, but the solvation energy is -0.7 eV in LiCl⁵³ and -1.0 eV in LiI.⁵⁴

4. SUMMARY

The computations in this article show that solvated electrons in molten salts are powerful reducing agents and are likely to be interesting base catalysts. In principle, these electron-in-molten-salt catalysts have a number of advantages: (1) The chemical potential of the solvated electron can be tuned by changing the molten salt. In the case of molten halides, which are of interest here, the melt may contain any alkali cation

combined with any halogen ion, and the melt could be a pure halide or a mixture of halides. One can dissolve in such a melt any alkali metal. (2) Molten salts containing solvated electrons can be used in a wide temperature range because the salts make eutectics that melt at low temperature, and they can be heated to high temperature because they have low vapor pressure. (3) Molten catalysts do not coarsen, are much easier to prepare than the customary heterogeneous catalysts, and are less susceptible to poisoning.^{55–57} (4) Many important catalyzed reactions are plagued by coking. This is not a problem when using a molten salt catalyst because the carbon floats to the surface of the melt where it could be skimmed off. Traditional solid catalysts are deactivated by coke. (5) The high heat conductivity and heat capacity of the molten salts are helpful for heat management. (6) Solvated electrons are produced in the electrochemical industrial process for the preparation of Li or Na, and understanding their properties or possible use is of interest. (7) One can envision a “universal” method of preparing clusters of a metal M, in a molten alkali halide salt, by dissolving the halide of M in the melt and reducing it by adding an alkali metal to provide solvated electrons. If the amount of alkali metal exceeds the amount needed for the reduction of the halide of M, one can charge the metallic clusters negatively. Such charging is often invoked when explaining the role of alkali metal promoters; prominent examples are the Fe²⁰ and Ru catalysts²¹ for ammonia synthesis. (8) The simplest way to use a solvated electron catalyst is to bubble the gaseous substrate through a molten salt bed containing solvated electrons. If the surface of the bubbles is too small, one can use a supported molten alkali halide in which one dissolves an alkali metal. Supported molten halides have been used in a fixed-bed reactor for oxidative ethane dehydrogenation;^{58,59} supported molten CuCl₂ has been used as a Deacon catalyst;⁶⁰ and supported vanadium oxide dissolved in molten alkali pyrosulfate is the catalyst for SO₂ oxidation in sulfuric acid production.⁶¹ It is therefore possible to envision a fixed-bed catalytic system consisting of a supported molten halide in which an alkali metal is dissolved.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b05716.

Illustration of bipolaron movement in the 2Li⁺ + 2e[−] + 45LiCl system, evidence against the formation of Li clusters in Li/LiCl, details concerning Na solvation in NaCl, evidence against the formation of Na₂ clusters in Na/NaCl, details concerning reactions between AgCl and Na/NaCl, ionic mixing in LiCl–NaCl, distribution of cations around solutes in different molten salts, and enthalpy of solid Ag at 900 K (PDF)

Movement of a bipolaron (MPG)

Movement of ions in the CH₄ + 2Li + 45LiCl system (MPG)

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Notes

The authors declare no competing financial interest.

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